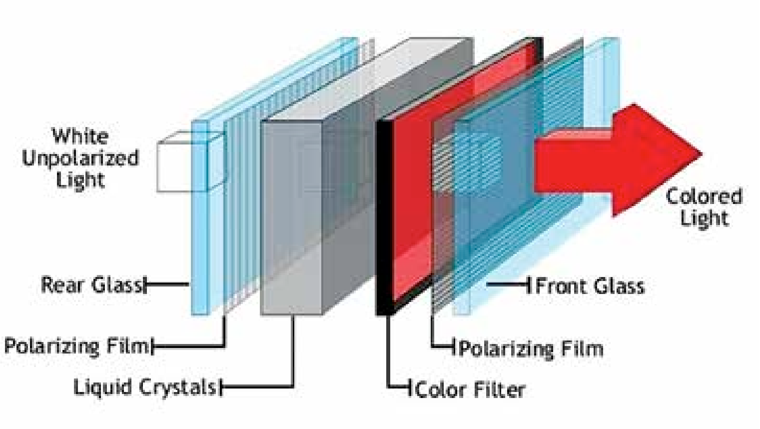
Light emitting diodes (LEDs)

Jumping on to more recent and widely used technology, we have LED screens, which work similar to LCDs. You still have your multiples layer including two polarisers and a layer of liquid crystals. The only real difference lies in using LEDs to backlight the screen instead of white fluorescent tubes, as in LCDs. After the discovery of white light LEDs, their use in display screens became more widespread and led to the development of the LED screen as we know it today. However, unlike in an LCD, the LED can be placed at various positions behind the screen



Cross section of a typical LED panel

to change effectiveness and viewing angles. You have your edge lit LED screens, with a row of LEDs around the rim of the screen, as seen commonly in mobile phone displays. Most large screen TVs and displays use the full array backlighting, with a full layer of LEDs behind the screen. However, this forces one to give up control of individual LEDs. That was offset by the introduction of dynamic backlights, where individual LEDs could be controlled to dim the screen in certain spots, lending even higher contrast to the pictures.

Many sets use white LED edge lighting to shine light across the display. RGB LED-backlit sets, on the other hand, provide improved color. However, since the technology still uses the liquid crystals of LCDs, there is no real advantage apart from the weight and thickness of the screen. Hence, the main innovation lies in the use of OLEDs, which we cover next.